

Marketing Channel Removal Effect Analysis

Markov Chain Based Marketing Attribution Using 15 Million User Path Logs

1. Objective

The objective of this project is to analyze 15 million user path logs and identify the contribution of individual marketing channels toward final conversions. A transition probability matrix is constructed from the user journeys, and the Removal Effect is calculated by systematically removing each channel and measuring the resulting reduction in conversion probability.

2. Input Data

Each user journey consists of a sequence of marketing touchpoints followed by either a conversion or non-conversion outcome.

Example: Google Ads → Facebook → Email → Conversion
Facebook → Instagram → Conversion
Google Ads → Email → Conversion

3. Transition Probability Matrix

For every consecutive pair of states in the user paths, transition counts are calculated. These counts are normalized to obtain transition probabilities.

P_{ij} = Number of transitions from state i to state j / Total transitions from state i

From / To	Google	Facebook	Email	Conversion
Google	0.00	0.40	0.30	0.30
Facebook	0.20	0.00	0.50	0.30
Email	0.10	0.20	0.00	0.70

The example matrix is illustrative; the actual matrix must be calculated from the 15 million logs.

4. Conversion Probability

After constructing the transition matrix, calculate the probability that a user starting from the initial state will eventually reach the Conversion state. This value represents the baseline conversion probability before removing any marketing channel.

Baseline Conversion Probability = $P(C)$

5. Removal Effect

For each marketing channel, remove the corresponding node from the transition graph, redistribute the affected transition probabilities appropriately, and recalculate the probability of reaching Conversion.

Removal Effect $_{i,j}$ = $[P(C) - P(C \mid \text{channel } i \text{ removed})] / P(C)$

A higher Removal Effect indicates that the channel has a greater influence on the overall conversion probability.

6. Example Calculation

Assume the baseline conversion probability is 0.40. After removing Google Ads, the conversion probability becomes 0.28.

$$\text{Removal Effect} = (0.40 - 0.28) / 0.40 = 0.30 = 30\%$$

7. Expected Output

Marketing Channel	Removal Effect
Google Ads	30%
Facebook	24%
Email	18%
Instagram	14%
YouTube	9%

The values above are illustrative and demonstrate the expected final ranking format.

8. Technology Stack

Python, Pandas, NumPy, SciPy, PySpark, Matplotlib, and Plotly can be used. PySpark is particularly suitable when processing 15 million user path records at scale.

9. Project Workflow

15 Million User Path Logs → Data Cleaning → Extract Channel Transitions → Build Transition Count Matrix → Convert Counts to Probabilities → Calculate Baseline Conversion Probability → Remove Each Channel → Recalculate Conversion Probability → Calculate Removal Effect → Rank Channels → Visualization and Report

10. Conclusion

This project provides a mathematical approach for marketing attribution using a Markov-chain transition model. By measuring the relative reduction in conversion probability after removing each channel, organizations can identify the marketing touchpoints that contribute most strongly to conversions and use the results to support marketing budget and campaign decisions.